

MAJOR PROJECT DISSERTATION
ON
“EXPLAINABLE AI:
DETECTING CLIMATE SIGNALS WITH
LAYER-WISE RELEVANCE PROPAGATION”

IN PARTIAL FULFILLMENT OF
B.Tech. (AIDS)
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DECLARATION BY THE CANDIDATE

I hereby declare that the work, which is being presented in this project entitled **“Explainable AI: Detecting Climate Signals with Layer-Wise Relevance Propagation”**, is an authentic record of my own work carried out by me under the supervision and guidance of **Mr. Anupam Lakhanpal, Assistant Professor, University School of Automation and Robotics, GGSIPU.**

This project was undertaken as a part of the major project report for partial fulfillment of B.Tech. (AIDS).

I have not submitted the matter embodied here in this project for the award of any other degree/diploma.

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Branch: AIDS

CERTIFICATE

This is to certify that this MAJOR PROJECT REPORT “**Explainable AI: Detecting Climate Signals with Layer-Wise Relevance Propagation**” is submitted by **Shreya Shree** who carried out the project work under my supervision. I approve this project for submission of the Bachelor of Technology (B.Tech, AIDS) in the University School of Automation and Robotics (USAR), Guru Gobind Singh Indraprastha University, Delhi.

Date: 12-05-2026

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Thanking You

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ABSTRACT

Climate change is one of the major global challenges, and understanding the causes behind regional temperature variations is an important area of research. Factors such as greenhouse gases and industrial aerosols significantly influence climate patterns, but separating their individual effects remains difficult. Modern deep learning models can analyze large amounts of climate data effectively, yet they often lack transparency in explaining how predictions are generated.

To address this issue, this project uses an Explainable Artificial Intelligence (XAI) technique known as Layer-wise Relevance Propagation (LRP). LRP is designed to interpret neural network predictions by identifying which input features contribute the most to the final output. Instead of treating the model as a black box, LRP helps visualize the decision-making process by propagating relevance values backward through the network layers.

The study uses climate data generated from the Community Earth System Model (CESM) large ensemble simulations. These simulations are performed under different combinations of greenhouse gas and aerosol forcing conditions over several decades. Annual maps of near-surface temperature are collected from these simulations and used as input data for training a neural network model.

The neural network is trained to predict the corresponding year of each climate temperature map. Once the model achieves accurate predictions, LRP is applied to generate relevance maps that highlight the regions and temperature signals most important for the predictions. These relevance maps help in understanding which geographical areas are strongly influenced by different climate forcings.

The proposed approach combines deep learning and explainable AI to improve both prediction performance and interpretability in climate studies. By identifying the most influential regions and patterns, this project provides better insight into the effects of greenhouse gases and aerosols on regional climate behavior. The study can contribute to future research in Earth sciences by making climate prediction models more understandable and reliable.

Keywords: Layer-wise Relevance Propagation (LRP), Explainable AI, Relevance Maps, Neural Networks, Climate Change, Large Ensemble Simulations

LIST OF ABBREVIATIONS

1. ANN – Artificial Neural Network
2. LRP – Layer-wise Relevance Propagation
3. XAI – Explainable Artificial Intelligence
4. GCM – General Circulation Model
5. CESM – Community Earth System Model
6. LENS – Large Ensemble (CESM Large Ensemble Project)
7. ALL – Simulations with all forcings (combined greenhouse gases, aerosols, and others)
8. GHG+ – Greenhouse Gas forcing simulation data
9. AER+ – Aerosol forcing simulation data
10. 20CRv3 – Twentieth Century Reanalysis Version

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ROLES & RESPONSIBILITIES

From the beginning of the project, both team members were meant to work with their full capacity in various phases during the making of this project. As time progressed, some roles and responsibilities were divided between the members while ensuring a full grasp of each other's work.

Shreya Shree (11819051922) undertook the responsibility of implementing the ANNs and LRP algorithm that serve as the heart of this project. She also took the lead in communicating and reporting progress to the mentor.

Ayush Kumar (11919051922) was assigned the responsibility of designing the pipeline of this project, along with handling data operations (collection and preprocessing). He also took the role of finding results and documenting the findings.

CHAPTER 1: INTRODUCTION

1.1 Background

Machine learning models are playing a really transformative role in Earth science and many more related applications. They enable the capture of complex non-linear relationships, help in the analysis of spatially distributed data, and typically require significantly less computational resources compared to traditional General Circulation Models (GCMs), as GCMs work on real physical laws that require huge number of parameters, rather than learning pattern of data.

Recently, machine learning models have been used for diverse applications in mesoscale meteorology, numerical weather prediction, simulating cloud and radiation processes in GCMs, turbulence and convection parameterizations, attribution of global climate change, and reconstructions of historical temperature trends.

However, despite these advantages, it remains challenging to disentangle the relative influences of aerosols and greenhouse gases on regional surface temperature trends within the broader context of global climate change. The intricate interactions between internal variability and external climate forcings further complicate the interpretation of the underlying physical mechanisms that drive regional, and even global-scale, temperature variability. The adoption of machine learning applications for geoscience issues continues to rapidly grow, especially due to an increasing number of XAI methods.

To explore how machine learning models are making their predictions, we focus on using XAI techniques in order to gain new scientific insights for climate science. Explainable AI (XAI) looks at the problem in the given direction: First, a model is trained without caring too much about feature selection. Only after training we look at which input features the neural network has learned.

Among different XAI approaches, Layer-wise Relevance Propagation (LRP) has emerged as an effective method for interpreting deep learning models. LRP traces the prediction process backward through the neural network and assigns relevance scores to input features that contribute most strongly to the output. In climate science, this helps researchers visualize important geographical regions and temperature patterns influencing model predictions. By improving transparency and interpretability, LRP

supports a deeper understanding of climate forcing mechanisms and strengthens trust in AI-based climate analysis systems.

1.2 Objectives of the Project

In this project, Artificial Neural Networks (ANNs) are trained in with an explainability technique known as Layer-wise Relevance Propagation (LRP) to analyze data derived from climate model simulations. By comparing LRP outputs across different ANN configurations, we aim to identify and interpret climate patterns associated with changing combinations of external forcings, specifically greenhouse gases and industrial aerosols.

Specifically, the model aims to learn the relationship between spatial patterns of near-surface temperature and their corresponding temporal information, represented in terms of years or decades. By training the ANN on multiple ensemble simulations, our objective is to capture robust and useful climate signals while accounting for internal variability present in the climate system.

Through LRP, the project aims to create relevance heatmaps and identify the spatial regions that contribute most significantly to the model's predictions, and thus link learned patterns to physically meaningful climate processes such as regional or global warming trends. Also, the project aims to find how different preprocessing and modeling choices influence both prediction accuracy and interpretability.

Another important objective of this project is to improve the transparency of deep learning models used in climate studies. Many neural network models can provide accurate predictions, but understanding why a particular prediction is made is often difficult. By using LRP, this project aims to make the prediction process more understandable for researchers by clearly showing which climate regions and temperature patterns are most influential in the model's decision-making process.

The project also aims to demonstrate how Explainable AI techniques can support future climate research and environmental studies. The generated relevance maps can help scientists better understand the effects of different climate forcings on regional temperature changes. In addition, this work encourages the use of interpretable AI

models in Earth science applications, where both prediction accuracy and scientific understanding are equally important.

1.3 Problem Statement

Despite the growing success of machine learning models in climate science, their application is often constrained by limited interpretability and a lack of physical transparency. While neural networks are capable of learning complex, nonlinear mappings between high-dimensional spatial temperature fields and their temporal evolution, they typically function as black-box models. This makes it difficult to discern the underlying decision-making process, particularly in terms of identifying salient spatial features and understanding the contribution of different regions to the model output. Such limitations hinder the integration of data-driven approaches with established physical understanding, thereby restricting their utility in scientific inference.

Moreover, the climate system is governed by intricate interactions between anthropogenic forcings, such as greenhouse gases and aerosols, and intrinsic internal variability, leading to spatiotemporally heterogeneous temperature responses. The superposition of these signals introduces significant challenges in attributing regional climate trends and disentangling competing influences. Traditional statistical methods often lack the capacity to capture these high-dimensional, nonlinear dependencies in a coherent framework. Therefore, there exists a need for a robust modelling approach that not only captures the spatiotemporal dynamics of climate variability but also facilitates interpretability through principled attribution techniques. This project addresses this gap by employing an Artificial Neural Network to model the underlying relationships and leveraging Layer-wise Relevance Propagation (LRP) to extract physically meaningful insights from the learned representations.

Another major problem in climate-related machine learning studies is the difficulty in validating whether the learned patterns are scientifically meaningful or simply statistical correlations. Since climate datasets contain large amounts of spatial and temporal information, neural networks may sometimes focus on irrelevant features that do not represent actual climate processes. Without proper interpretability methods, researchers cannot confidently determine whether the model predictions are physically reliable and useful for scientific analysis.

CHAPTER 2: LITERATURE REVIEW

With the increasing use of AI systems in different fields, the need for models that are easier to understand has also increased. In addition, the ability to understand how AI models make decisions has become important for improving reliability and user trust. Explainable artificial intelligence (XAI) focuses on methods that help users understand how machine learning models generate predictions [1]. However, trustworthy XAI systems become reliable explainability methods that are necessary for the practical use of AI systems [2].

On the topic of applicability and versatility, XAI techniques have equipped developers, domain experts, and decision-makers with valuable tools to understand the working process of complex machine learning models using tabular data, images, language, and graphs [3]. By delving into the implementation details, we aimed to make the internal behavior of these models easier to understand, fostering a didactic environment for users seeking clarity.

Machine learning is now widely used in Earth science because it can process large climate and environmental datasets efficiently. Both classical ML (e.g. regression, Gaussian processes) and deep learning are used for tasks ranging from weather forecasting and climate pattern detection to computational emulation of complex simulation outputs. For example, a neural network trained on global sea-surface temperature fields can distinguish El Niño vs La Niña events with near-perfect accuracy, demonstrating that it has captured the characteristic tropical Pacific patterns of the phenomenon. Recent reviews note that deep learning can automatically extract spatio-temporal features to improve process understanding and predictive skill. These advances offer powerful new capabilities for climate simulation and prediction, but they also raise concerns about interpretability and physical consistency.

The geoscience community has increasingly turned to explainable AI (XAI) methods. XAI methods help explain how machine learning models make decisions. In climate and weather science, explainability is valued both for validating that a model learns physically plausible relationships and for increasing confidence in AI-based systems. XAI methods can improve confidence and encourage wider use of AI methods in geoscience by making decision processes transparent [4]. They observe, however, that uptake of XAI has been limited: only a small fraction of geoscience AI studies employ

explainability tools. Nevertheless, a growing number of studies apply XAI algorithms (e.g. saliency maps, concept relevance, feature attribution) to problems such as climate signal detection and extreme-event prediction. These methods help determine whether a model is focusing on sensible geophysical features (for example, whether a weather prediction model relies on the correct large-scale circulation patterns).

A prominent XAI approach for neural networks is Layer-wise Relevance Propagation (LRP). Introduced by [5] for deep image classifiers, LRP works by tracing the prediction backward through the neural network so that each input dimension (e.g. each pixel) is assigned a “relevance” value. The result is a relevance heatmap that shows which inputs *positively* contributed to the output. As Bach *et al.* describe, these the results are displayed as heatmaps showing important regions that allow a human to “verify the validity of the classification decision”.

In geoscience applications, LRP has been used to expose learned patterns: for example, , global LRP is a powerful tool for understanding regression predictions made by the CNN models applied to problems in the Earth sciences and highlight the benefit that this method provides in showing the nonlinear and nonlocal effects of the model predictors [6].

In the context of generally using images as input, [7] focused on the task of visualizing important regions in an image as a visual explanation of the model’s decisions. They extended LRP to models with residual connections such as ResNet by introducing Relevance Splitting at points where the output from a skip connection converges with that from a residual block and their work demonstrated superior performance to that of baseline methods. While dealing with problem of interpretability with an LSTM, [8] developed a feature-level spatial–temporal layer-wise relevance propagation (ST-LRP) method with a firm explanation by improving layer-wise relevance propagation (LRP). Their method quantitatively obtains the correlation of multiple input feature data to the energy consumption prediction results in both spatial and temporal dimensions.

Unsupervised learning algorithms extract structures from unlabeled data from which properties such as membership to some cluster or degree of anomaly can be predicted. In order to explain these predictions, a novel methodology called Neuralization-Propagation (NEON), the learned unsupervised model is first ‘neuralized’ (i.e.

transformed into a functionally equivalent neural network). Then, an LRP procedure is built in order to propagate the prediction backward in the neural network [9].

LRP was also applied to extract insights from well-functioning ML models, e.g. in face expression recognition [10]. LRP was used to find relevant features for audio source localization [11], and to identify EEG patterns that explain decisions in brain-computer interfaces [12].

In recent years, researchers have also focused on improving the transparency of neural network predictions through explainable AI methods. Traditional deep learning models often provide high prediction accuracy but limited interpretability, which reduces trust in scientific applications. Explainability methods such as saliency mapping, gradient-based attribution, and Layer-wise Relevance Propagation (LRP) are now widely explored for understanding the internal decision-making behavior of deep learning systems. These methods help identify the most influential input regions responsible for model predictions.

CHAPTER 3: METHODOLOGY

3.1 Proposed Design Flow

The plan is summarized in the following flow chart-

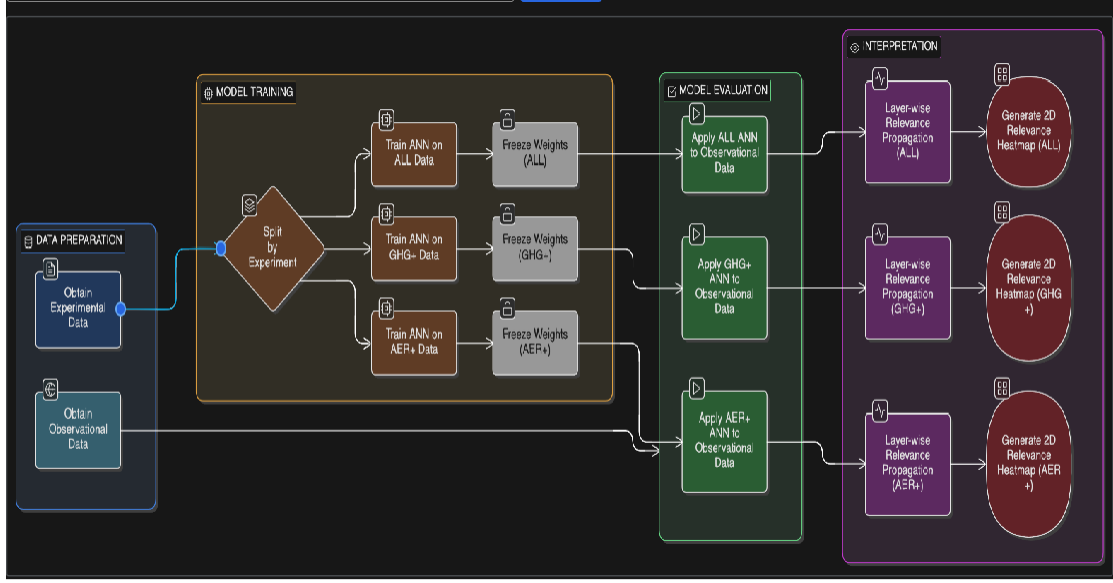


Figure 3.1. Summarization of the project

There are three experiments (all fields are monthly 2 m air temperature, output on a 1° CAM5 atmosphere, then regridded to 1.9°×2.5°):

1. ALL (All forcings): 20 members with historical forcing (1920-2005) followed by RCP8.5 (2006–2080), including time-varying greenhouse gases, aerosols, land-use, ozone, etc.
2. GHG+ (Fixed aerosol): 20-member ensemble with greenhouse gases evolving as above, but industrial aerosols held fixed at 1920 levels.
3. AER+ (Fixed GHG): 20-member ensemble with aerosols evolving but greenhouse gases fixed to 1920 levels.

Figure 2 shows annual maps of temperature trends over four separate 40-year periods for the ensemble mean of each experiment.

For observations, we use the NOAA/CIRES/DOE 20th Century Reanalysis version 3 (20CRv3), covering 1920–2015, with 2 m air temperature on a global grid. The reanalysis fields (monthly 2 m temperature) are interpolated to the same 1.9°×2.5° grid. Data and subsequent actions on it are described in Section 3.2.

We then utilize layer-wise relevance propagation (LRP) to visualize the regional temperature signals that are important for the ANN's prediction in each climate model experiment.

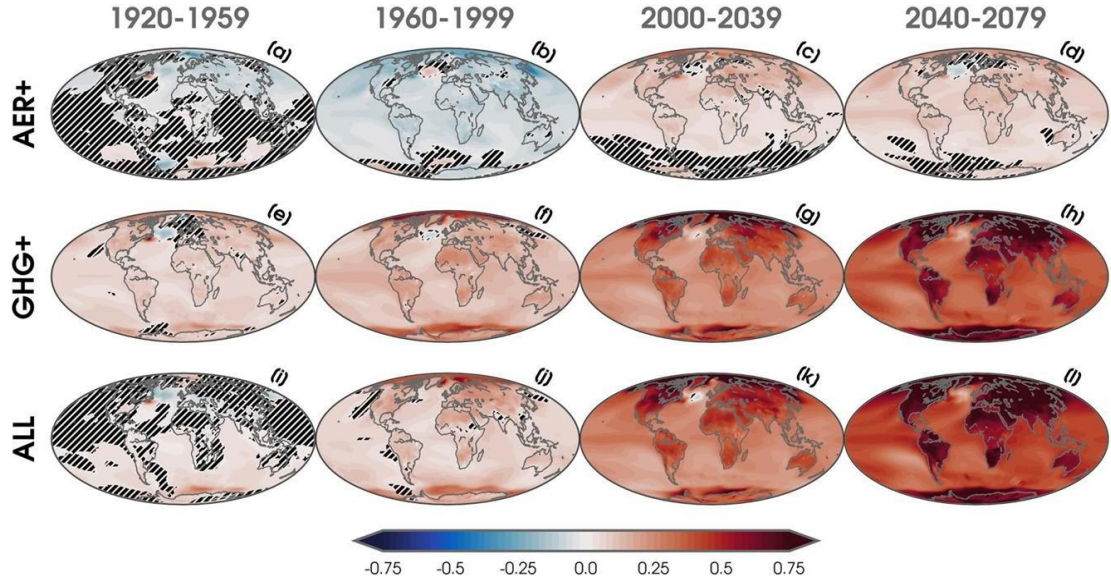


Figure 3.2. Annual linear least squares trends of 2-m temperature (°C per decade) over 1920–1959 (a, e, i), 1960–1999 (b, f, j), 2000–2039 (c, g, k), and 2040–2079 (d, h, l) for the ensemble means of three climate model simulations (AER+; a–d, GHG+; e–h, ALL; i–l). Statistically significant trends are shown with shaded contours at the 95% confidence level following the Mann-Kendall test, while those that are not are masked out using black hatch marks.

Figure 2 presents the spatial temperature trends for different time periods under three climate forcing experiments: AER+, GHG+, and ALL. The maps show annual trends in near-surface temperature from 1920 to 2079, divided into four separate periods: 1920–1959, 1960–1999, 2000–2039, and 2040–2079.

In the AER+ experiment, temperature changes remain relatively weak during the earlier periods, with some regions showing cooling effects. Even in later periods, warming is present but comparatively smaller than in the other experiments. This behavior suggests the cooling influence of aerosols on global temperature.

The GHG+ experiment shows a clear and continuous increase in warming across almost all regions of the world. The warming becomes much stronger after the year 2000, especially over land areas and polar regions. By the period 2040–2079, strong warming patterns are visible globally, highlighting the impact of greenhouse gas emissions on long-term climate change.

The ALL experiment combines the effects of both greenhouse gases and aerosols. In the earlier decades, some cooling regions are visible because of aerosol influence, but over time the greenhouse gas effect becomes dominant. In the future periods, the maps show widespread warming similar to the GHG+ experiment, indicating that greenhouse gas forcing has a stronger long-term impact on global temperature trends.

These temperature trend maps help visualize how different climate forcings influence regional and global warming patterns over time.

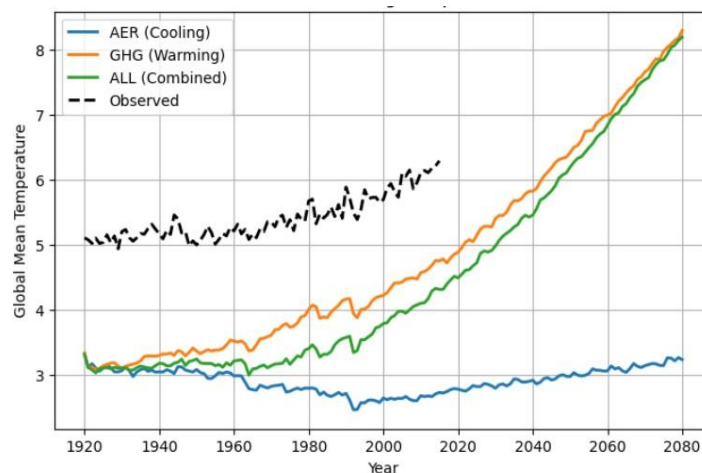


Figure 3.3. Temporal trends of 2-m temperature ($^{\circ}\text{C}$ per decade) over 1920-2080 for the annual ensemble means of three climate model simulations (AER+, GHG+, ALL) with solid lines, and annual means of the Twentieth Century Reanalysis version 3 observations over 1920-2015 shown with dashed line.

Figure 3 shows the change in global mean near-surface temperature from 1920 to 2080 for different climate forcing experiments..

The GHG+ simulation shows a strong increase in temperature over time, indicating the warming effect caused by greenhouse gases. The ALL simulation also shows a significant rise in temperature because it includes the combined influence of all climate forcings. In contrast, the AER+ simulation displays a much weaker warming trend, suggesting that aerosols have a cooling influence on the climate system.

The observed temperature trend follows a pattern similar to the ALL simulation, especially in later decades, which suggests that the combined forcing experiment better represents real-world climate behavior. The graph also shows that temperature rise becomes more rapid after the late twentieth century, highlighting the increasing impact of human-induced climate change over time.

3.2 Proposed Methodology

For all climate model data, we use large ensemble simulations performed by the Community Earth System Model version covering 1920 to 2080. The atmospheric model is the Community Atmosphere Model version 5, which is coupled to interactive land, ocean, and sea ice components.

Within a single large ensemble GCM simulation, one can obtain the forced response (i.e., climate signal) by averaging across individual ensemble members that differ by only a small random perturbation error. Thus, if the model is correct, observations of the real world should fall within the ensemble spread in order to reflect both a common forced signal (climate change) and the unpredictable noise of the atmosphere

Here, we first analyze the widely used 20-member large ensemble, which we refer to as “ALL” (for all-forcing). The large number of ensemble members is useful for characterizing atmospheric internal variability (or noise) in the climate system. Each of the ensemble members have the same external forcing, but are generated from a small random round-off difference in the atmospheric initial conditions. Historical forcing is imposed from 1920 to 2005, and thereafter Representative Concentration Pathway 8.5 is used to simulate a worst-case climate scenario through the end of the 21st century. Land use/land cover changes, biomass burning, and stratospheric ozone concentrations also evolve with time in the ALL simulation.

In addition, we also use a set of two new single-forcing simulations from CESM1 that are both also run with 20 ensemble members. These large ensembles have the same GCM, initialization protocol, and external forcing as ALL, but differ by one time-evolving forcing agent that is withheld per simulation. In particular, greenhouse gas concentrations are held fixed to 1920 levels in one experiment (AER+), and industrial aerosols are held fixed to 1920 levels in another (GHG+). We are going to call these “AER+” and “GHG+”, respectively.

We apply a bilinear interpolation to the three sets of large ensembles so that they share a slightly coarser latitude by longitude global grid ($1.9^\circ \times 2.5^\circ$). We only consider fields of monthly near-surface air temperature to calculate annual means from model output.

Before training the Artificial Neural Network (ANN), the climate datasets undergo several preprocessing steps to ensure consistency and improve model performance. Since the datasets are collected from different ensemble simulations and observational sources, all temperature fields are converted into a common spatial resolution using bilinear interpolation. This allows the ANN to process data with uniform dimensions across all forcing experiments.

The monthly near-surface temperature data is then converted into annual mean temperature maps by averaging all twelve months of each year. This helps reduce short-term fluctuations and allows the model to focus on long-term climate trends. After generating annual maps, each dataset is normalized using mean subtraction and standard deviation scaling. Normalization improves training stability and prevents regions with larger temperature values from dominating the learning process.

The processed temperature maps are then flattened into one-dimensional vectors before being provided as input to the ANN model. Each input sample therefore represents the global spatial temperature distribution for a particular year and ensemble member.

To evaluate the learning capability of the proposed ANN models, the available ensemble members are divided into training and validation datasets. Out of the 20 ensemble members, 16 members are used for training while the remaining 4 members are reserved for validation and testing. This separation helps ensure that the model is evaluated on unseen climate data and does not simply memorize the training samples.

During training, the ANN learns the relationship between spatial temperature patterns and their corresponding temporal information. The network weights are updated iteratively using backpropagation and gradient-based optimization techniques. Multiple epochs are used to improve learning performance and minimize prediction error.

Validation is performed after training to measure how accurately the model can predict years from previously unseen temperature maps.

3.2.1 Algorithm Formulated

3.2.1.1 ANN Training and Testing:

We utilize Artificial Neural Networks (ANNs) for analysis. We compare the impact of time-evolving greenhouse gases and industrial aerosols on a classification task of predicting the decade (year) from input maps of temperature. Each unit of the ANN input layer represents one grid point from a 2-m temperature map (13,824 units per map with dimensions of 96 latitudes by 144 longitudes), and our output layer represents the probabilities of a particular decade class.

We set up two ANN architectures. The network in first architecture has two hidden layers with 20 neurons each . All hidden-layer activations use Rectified Linear Units (ReLU), accompanied with Batch Normalization and Dropout. The second architecture contains three layers with number of units as 128, 64, and 32.

The output layer uses a softmax activation over decadal classes (e.g. 1920–1929, 1930–1939, ...); in effect each output neuron represents the probability that the input map falls in that decade . This “fuzzy classification” is decoded into a continuous year by taking the weighted sum of the decade centers (for example, if the 2000–2009 class center is 2005 with probability 0.7 and 2010–2019 center is 2015 with probability 0.3, the predicted year is $0.7 \times 2005 + 0.3 \times 2015 = 2008$).

Each ANN is trained separately on data from one experiment (ALL, GHG+, or AER+). For a given experiment, we randomly select 80% of the ensemble members (16 members) for training and reserve the remaining 20% (4 members) for testing . The training optimizes a categorical cross-entropy (log-loss) objective using stochastic gradient descent with Nesterov momentum (momentum=0.9) . Hyperparameters are: learning rate = 0.01, batch size = 32, for 500 epochs . To prevent overfitting, an L2 (ridge) weight penalty of 0.01 is applied on the weights of the first hidden layer . These choices were verified to be robust (the results do not sensitively depend on slight changes in epochs or regularization).

To evaluate against real-world data, the trained ANN is applied to the observational reanalysis maps (20CRv3). Each annual mean map of observed 2 m temperature is input to the network (standardized using the same scalings as the model data). The ANN produces a predicted year, which is compared to the true year.

```

#20*20
import torch.nn as nn

input_dim = X_train_tensor.shape[1]

model = nn.Sequential(
    nn.Linear(input_dim, 20),
    # nn.BatchNorm1d(20),
    nn.ReLU(),
    # nn.Dropout(0.2),

    nn.Linear(20, 20),
    # nn.BatchNorm1d(20),
    nn.ReLU(),
    # nn.Dropout(0.2),

    nn.Linear(20, len(decade_centers))
).to(device)

```

Figure 3.4. Architecture of ANN with Two Hidden Layers (20–20 Configuration)

This architecture represents a fully connected ANN which is designed for detecting or learning climate pattern from near-surface temperature maps. This architecture consists of an input layer which has 2 hidden layers. Each hidden layer contains 20 neuron that is forming the 20-20 architecture.

The input dimension is determined dynamically from the shape of the training dataset, where each input sample corresponds to a flattened global temperature field. The hidden layers use the Rectified Linear Unit (ReLU) activation function to introduce non-linearity and enable the network to learn complex spatial relationships present in climate data.

The final output layer contains neurons equal to the number of decadal classes and produces prediction scores corresponding to different temporal categories.

This smaller ANN architecture is computationally efficient and suitable for identifying large-scale climate trends while reducing the risk of overfitting. It also serves as a baseline model for comparison with deeper ANN configurations used later in the study.

This architecture represents a deeper ANN which is designed to evaluate the effect of increased model complexity on learning. In this architecture, it consists of three hidden layers which contains 128,64 and 32 neurons respectively.

The main purpose of using this deeper architecture is to compare its performance with the smaller 20–20 ANN model. By increasing the number of hidden layers and neurons,

we check whether the network can learn climate patterns more effectively and produce better predictions. The performance of both models is compared based on prediction accuracy, correlation values, learning capability, and the relevance patterns generated using Layer-wise Relevance Propagation (LRP). The results show that the deeper model performs better in distinguishing different climate forcing scenarios, especially in the ALL and GHG+ experiments, which suggests that it can capture more detailed climate-related features from the dataset.

3.2.2.2 LRP for Explainability:

Explainable machine learning looks at the problem of learning useful features in the following manner: First, a model is trained without caring too much about feature selection. Only after training we look at which input features the neural network has learned. Based on this explanatory feedback, ‘bad’ features can be removed and the model can be retrained on the cleaned data. A simple method, Taylor Decomposition, produces explanations by performing a Taylor expansion of the prediction $f(x)$ at some nearby reference point $\tilde{x}$:

$$f(x) = f(\tilde{x}) + \sum_{i=1}^d (x_i - \tilde{x}_i) \cdot [\nabla f(\tilde{x})]_i + \dots$$

First-order terms (elements of the sum) quantify the relevance of each input feature to the prediction and form the explanation. Although simple and straightforward, this method is unstable when applied to deep neural networks. The instability can be traced to various known shortcomings of deep neural network functions: Shattered gradients - While the function value $f(x)$ is generally accurate, the gradient of the function is noisy, and Adversarial examples - Some tiny perturbations of the input x can cause the function value $f(x)$ to change drastically.

LRP leverages the graph structure of the deep neural network to quickly and reliably compute explanations. LRP operates by propagating the prediction $f(x)$ backward in the neural network, by means of purposely designed local propagation rules.

The propagation procedure implemented by LRP is subject to a conservation property, where what has been received by a neuron must be redistributed to the lower layer in equal amount. This behavior is analogous to Kirchhoff’s conservation laws in electrical circuits. Let j and k be neurons at two consecutive layers of the neural network.

Propagating relevance scores $(R_k)_k$ at a given layer onto neurons of the lower layer is achieved by applying the rule:

$$R_j = \sum_k \frac{z_{jk}}{\sum_j z_{jk}} R_k.$$

The quantity z_{jk} models the extent to which neuron j has contributed to make neuron k relevant. The denominator serves to enforce the conservation property. The propagation procedure terminates once the input features have been reached. If using the rule above for all neurons in the network, it is easy to verify the layer-wise conservation property $\sum_j R_j = \sum_k R_k$, and by extension the global conservation property $\sum_i R_i = f(x)$. Although LRP

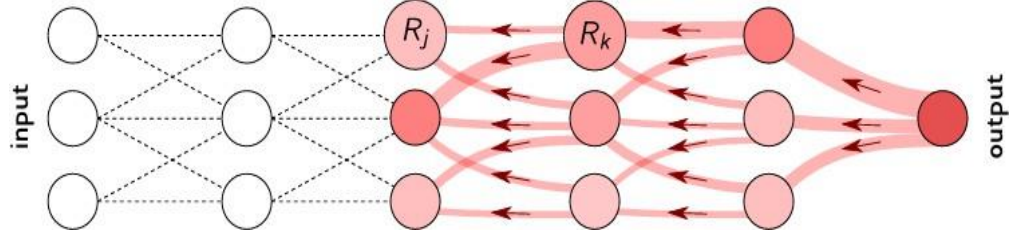


Figure 3.5. Illustration of the LRP procedure. Each neuron redistributes to the lower layer as much as it has received from the higher layer.

simple Taylor decomposition approach mentioned earlier, each step of the propagation procedure can be modeled as an own Taylor decomposition performed over local quantities in the graph.

To identify which regions drive the ANN’s predictions, we apply LRP . After training the network, the weights are frozen. For a given input sample (temperature map), the ANN’s highest-probability output neuron (the predicted decade class) is selected. LRP propagates this output backwards through the network to the input layer, following the “ $\alpha\beta$ -rule” suitable for ReLU networks . The result is a relevance score for each input neuron (grid cell) that indicates how much that pixel contributed to the prediction. In practice, one obtains a 2D relevance heatmap (same dimensions as the temperature map) for each sample.

After computing LRP scores for multiple input samples, the relevance values were aggregated by taking the mean relevance at each spatial grid point. This resulted in a single representative relevance map capturing the average contribution of each location.

To focus on the most significant features, only the top 10% of grid points with the highest absolute relevance values were retained. This thresholding based on absolute relevance ensures that both strongly positive and strongly negative contributions are considered, thereby highlighting regions that have the greatest influence on the model's predictions.

3.2.2 Mathematical Modelling:

We denote the raw climate data as spatial temperature fields $T_{t,m}(i, j)$, where t indexes year, m month, and (i, j) the latitude–longitude grid. We first aggregate monthly fields to annual means:

$$X_{t,e}(i, j) = \frac{1}{12} \sum_{m=1}^{12} T_{t,m,e}(i, j),$$

where e indexes the ensemble member. In this project, each climate scenario (e.g., “ALL”, “GHG+”, “AER+”) provides up to 20 ensemble members. We interpolate all fields (model and reanalysis) onto a common coarse grid of $1.9^\circ \times 2.5^\circ$ to ensure consistent input dimensions.

Each grid-cell time series $\{X_{t,e}(i, j)\}$ is then standardized to unit variance by dividing by its standard deviation $\sigma(i, j)$ and subtracted by its mean $\bar{X}(i, j)$ computed over all years t and ensemble members e . The network input is

$$X'_{t,e}(i, j) = (X_{t,e}(i, j) - \bar{X}(i, j)) / \sigma(i, j).$$

We split the available data across ensemble members 16 of 20 members (80%) are used for training and the remaining 4 for validation. Thus, we provide the ANN with inputs maps across 161 years and 16 members to learn from. Table 1 summarizes the symbols used.

Symbol	Definition
$T_{t,m,e}(i, j)$	2m temperature at year t , month m , member e , grid (i, j)
$X_{t,e}(i, j)$	Annual mean map at year t , member e : $= \frac{1}{12} \sum_m T_{t,m,e}(i, j)$.

Symbol	Definition
$X'_{t,e}(i, j)$	Normalized map: $(X_{t,e}(i, j) - \underline{X}(i, j))/\sigma(i, j)$.
m, n	Number of grid points in latitude, longitude (output resolution).
$f_\theta: R^{m \times n} \rightarrow R^D$	ANN with parameters θ . Input a map, output a D -vector of class logit scores.
D	Number of output classes (decades).
p_k	ANN softmax probability of class k . $\sum_{k=1}^D p_k = 1$.
y_k	Target probability for class k .
c_k	Central year of decade class k (e.g. 2005 for 2000–2009).
$\hat{t}$	Predicted year $\hat{t} = \sum_{k=1}^D p_k c_k$.
L	Loss (cross-entropy/KL divergence) between p and y .
MAE	Mean Absolute Error: $= \frac{1}{N} \sum_i t_i - \hat{t}_i $.

The ANN implements a nonlinear mapping f_θ from a flattened mn -dimensional input to D output units (one unit per decade). ReLU activations and a final softmax are used. We also use L2 weight regularization (ridge) on the first layer’s weights to suppress overfitting.

The output vector $z \in R^D$ is converted to a probability distribution by the softmax:

$$p_k = \frac{\exp(z_k)}{\sum_{l=1}^D \exp(z_l)}, \quad k = 1, \dots, D,$$

so that $\sum_k p_k = 1$. We define each class k to represent a decade with central year c_k (for example, $c_k = 2005$ for 2000–2009). During training, the true year t is encoded as a *fuzzy* target distribution y over decade-classes: typically one would set $y_k = 1$ if t falls in decade k , but we can allow a smooth encoding for ambiguity. The loss function is cross-entropy between the target *probabilities* and the network’s prediction.

In implementation, we minimize this loss using stochastic gradient descent. Each network is trained on 80% of the ensemble data (16 members) and remaining 20% (4 members) is used as validation data. After training, the estimated year for an input map

is found by the weighted average $\hat{t} = \sum_k p_k c_k$ of the decade centers (fuzzy decoding).

Layer-Wise Relevance Propagation:

LRP is applied to interpret which input pixels impact the ANN's prediction. Let $f(x)$ be the network's output score (logit or probability) for the predicted class. LRP propagates $f(x)$ back to the input features by a backward pass: each neuron redistributes its received *relevance* to the neurons in the lower layer. In the simplest “z-rule” variant, one sets

$$R_j = \sum_k \frac{z_{jk}}{\sum_{j'} z_{j'k}} R_k, \quad z_{jk} = a_j w_{jk},$$

where a_j is the activation of neuron j , w_{jk} the weight to neuron k in the next layer, and R_k the relevance at node k in the higher layer. Intuitively, z_{jk} measures how much j contributes to k 's activation. This rule enables conservation: one can show $\sum_j R_j = \sum_k R_k$ at each layer and hence $\sum_{i,j} R_{i,j} = f(x)$ for the input layer. In practice a small stabilizer (ϵ) is often added to denominators (LRP- ϵ rule) to avoid division by zero, but the fundamental idea is that relevance is apportioned proportionally to each input's contribution.

Consider the generic rule

$$R_j = \sum_k \frac{a_j \cdot \rho(w_{jk})}{\epsilon + \sum_{0,j} a_j \cdot \rho(w_{jk})} R_k,$$

of which LRP-0/ ϵ / γ are special cases. The computation of this propagation rule can be decomposed in four steps:

$$\begin{aligned} \forall_k : z_k &= \epsilon + \sum_{0,j} a_j \cdot \rho(w_{jk}) && \text{(forward pass)} \\ \forall_k : s_k &= R_k / z_k && \text{(element-wise division)} \\ \forall_j : c_j &= \sum_k \rho(w_{jk}) \cdot s_k && \text{(backward pass)} \\ \forall_j : R_j &= a_j c_j && \text{(element-wise product)} \end{aligned}$$

The first step is a forward pass on a copy of the layer where the weights and biases have been applied the map $\theta \rightarrow \rho(\theta)$, to which we further add the small increment ϵ . The second and fourth steps are simple element-wise operations. For the third step, one notes that c_j can also be expressed as the gradient computation:

$$c_j = \left[\nabla \left(\sum_k z_k(\mathbf{a}) \cdot s_k \right) \right]_j$$

where $\mathbf{a} = (a_j)_j$ is the vector of lower-layer activations, where z_k is a function of it, and where s_k is instead treated as constant

The outcome is a relevance map $R_{i,j}$ on the input grid: large positive R -values indicate grid cells that strongly support the predicted year. By construction $\sum_{i,j} R_{i,j} = f(x)$, so one can normalize the heatmap (for example, by its absolute sum) to compare across samples. In the climate context, these heatmaps highlight spatial temperature anomalies (e.g., Arctic warming) that drive the temporal prediction. Typically, we aggregate the pixelwise relevances across the test ensemble to find robust patterns.

3.3 Simulation Environment

The project was implemented in a Kaggle cloud notebook using freely available GPU resources (e.g. Tesla GPUs). The primary dataset consists of Community Earth System Model (CESM) initial-condition large-ensemble simulations under different forcing scenarios. Specifically, we use the CESM1 *single-forcing* ensembles where either industrial aerosols or greenhouse gases are held fixed (XAER and XGHG). Each ensemble has 20 members (1920–2080, CESM 1° grid), yielding multi-member time series of annual-mean near-surface temperature fields on a global grid (CESM1’s nominal 0.9°×1.25° grid, roughly 192×288 points). The data are uploaded to Kaggle as a dataset and accessed via the notebook’s input directory.

Within Kaggle’s environment, data files are organized under `../input/`, and custom code (Python 3) is developed in a Jupyter-like notebook. Kaggle notebooks come with standard ML libraries (e.g. TensorFlow, PyTorch) pre-installed and allow mounting additional datasets. An NVIDIA GPU (Tesla T4 or similar) is enabled to accelerate training.

The CESM1 Large Ensemble simulations used in this study are freely available ([Data Sets Available to the Community | Community Earth System Model](#)). Monthly 20th Century Reanalysis V3 (20CRv3) data are provided by the NOAA/OAR/ESRL PSL, Boulder, Colorado, USA, from their website at [Climate reanalysis | Copernicus](#).

The simulation environment used in this project provides a flexible and efficient platform for climate data analysis and deep learning model development. Kaggle's cloud-based notebook environment allows easy access to GPU resources, pre-installed machine learning libraries, and large datasets without requiring high-end local hardware. This setup helps in handling complex climate datasets, performing faster neural network training, and generating relevance maps effectively. The combination of publicly available CESM climate simulations and modern AI tools creates a reliable environment for conducting explainable climate modeling and temperature trend analysis.

CHAPTER 4: RESULTS & DISCUSSIONS

The results of this study are obtained by evaluating the performance of the proposed (ANN) models on unseen ensemble members and analyzing their predictive behavior in conjunction with interpretability techniques. The models are assessed based on their ability to accurately estimate the temporal information associated with spatial temperature maps, using metrics such as prediction error and bias.

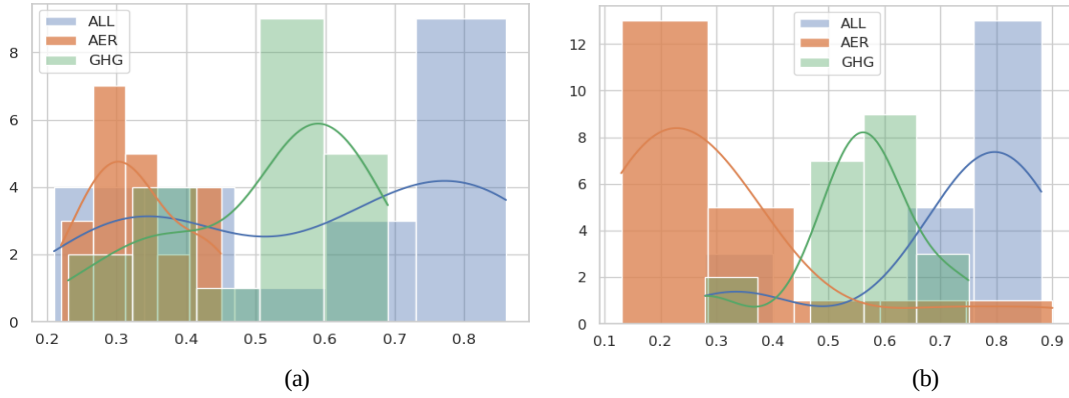


Figure 4.1. Histograms of the correlation between the actual years and the artificial neural network (ANN)-predicted years from Twentieth Century Reanalysis version 3 observations (a) Learning done by ANN with number of hidden units as (20,20) (b) Learning done by ANN with number of hidden units (128, 64, 32). The following results correspond to the smaller ANN, i.e., with hidden units (20, 20).

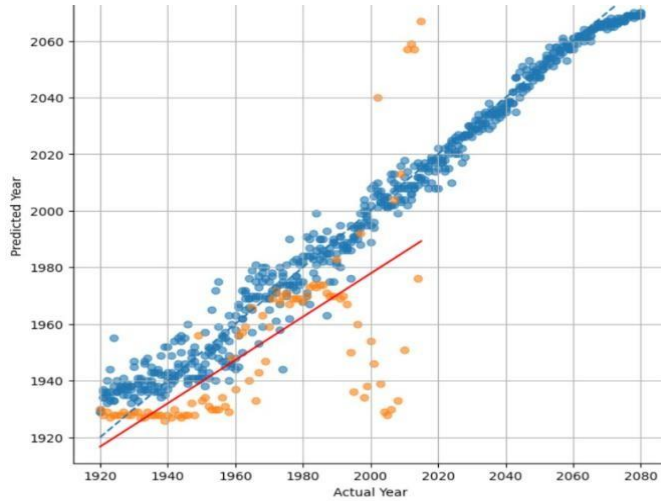


Figure 4.2. Scatter plot of predicted years versus actual years for the ANN model with two hidden layers (20–20), evaluated on 20CRv3 observational data for the period 1920–2015. Each point represents a single sample, with the solid line indicating the best linear fit to the predictions. The dashed line represents the ideal one-to-one relationship. The plot illustrates the correlation between predicted and observed years, along with any systematic deviation from perfect agreement.

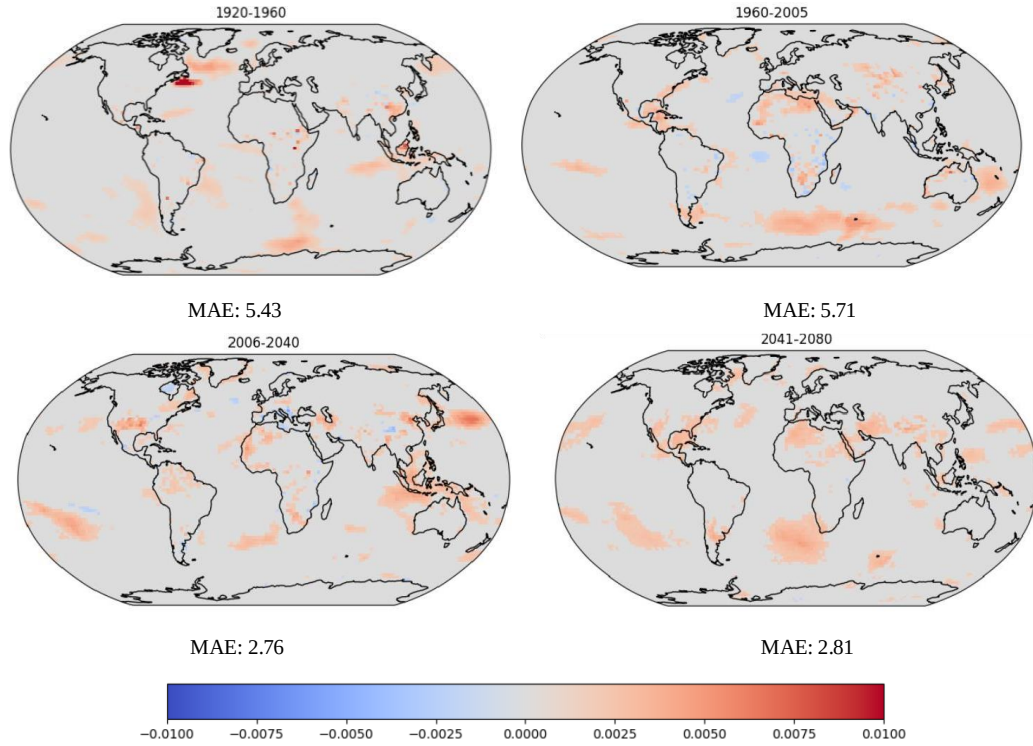


Figure 4.3. Layer-wise Relevance Propagation (LRP) relevance maps for four temporal periods learned by the ANN architecture with two hidden layers of 20 neurons each (20–20). The maps show the spatial distribution of relevance associated with the model’s year prediction for each period: 1920–1960, 1960–2005, 2006–2040, and 2041–2080. Warmer colors indicate positive relevance, while cooler colors indicate negative relevance. The mean absolute error (MAE), between ANN predictions and true values for validation data, for each period is also reported below the corresponding map.

The relevance scores are aggregated across multiple samples to obtain representative patterns, and a thresholding approach is used to highlight the most influential regions.

For both architectures, the ANN demonstrates a clear ability to learn the temporal evolution of temperature fields, as we can see the positive correlation values across all forcing scenarios. However, notable differences are observed in the distribution and spread of correlations between the two models. In the case of the smaller network, the correlations are moderately distributed, with GHG+ forcing showing relatively higher correlation values compared to AER+ forcing. The ALL forcing scenario results in a broader spread, indicating variability in model performance across samples.

In contrast, the deeper ANN architecture shows a more well-separated behavior between the forcing scenarios. The ALL simulations achieve consistently higher correlation values, that we may interpret as improved learning of combined forcing effects. The GHG+ simulations also give strong correlations, though slightly lower than ALL, reflecting the dominant influence of greenhouse gas forcing on long-term

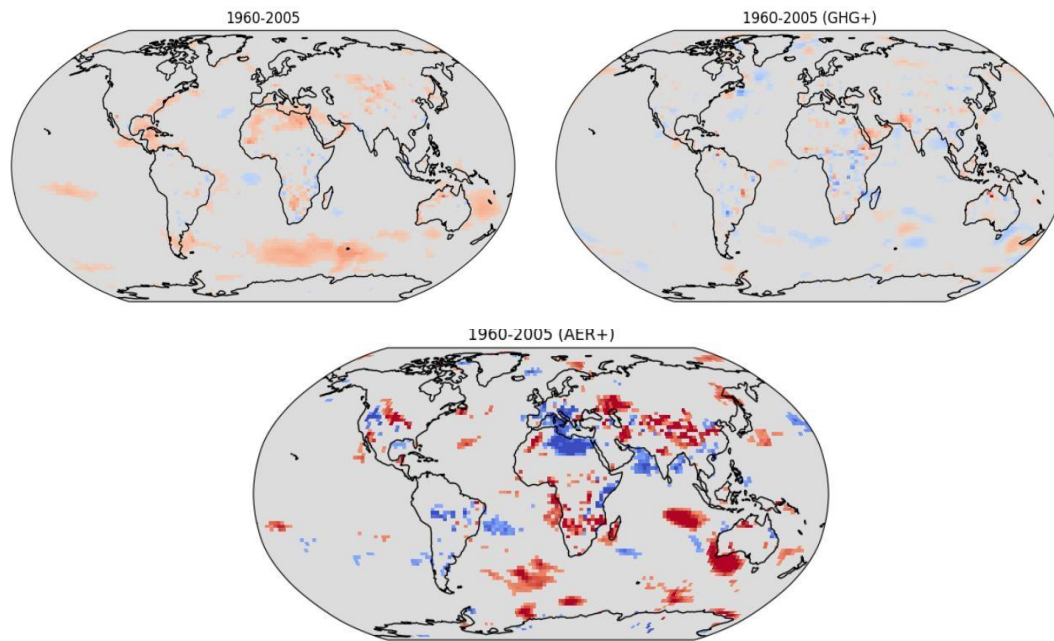


Figure 4.4. LRP relevance maps for the period 1960–2005 derived from the ANN (20–20) for three forcing experiments: ALL, GHG+, and AER+

temperature trends. On the other hand, the AER simulations show comparatively lower correlation values, suggesting that aerosol-driven temperature patterns are more complex and harder for the model to capture accurately.

From Figure 4.4, we find that, in the early period (1920–1960), relevance is relatively sparse but we can see some contributions over the North Atlantic and parts of the North Pacific, giving us a hint that the model relies on oceanic variability in these regions to distinguish earlier climate states.

In the periods that come later (2006–2040 and 2041–2080), the relevance becomes more widespread and spatially coherent, with regular contributions from the North Atlantic, and there are also enhanced signals over the Indo-Pacific and Southern Ocean regions. The impact of the North Atlantic across all time periods tells us that it plays a key role in the model’s ability to distinguish temporally, maybe it is due to its sensitivity to both internal variability and other external factors. Ofcourse, it is much out of our domain to confidently tell what geophysical laws and patterns are giving us these signals.

From Figure 4.4, the relevance maps for the period 1960–2005 do not give us a much coherent or big spatial structure across the three forcing experiments. In the ALL and

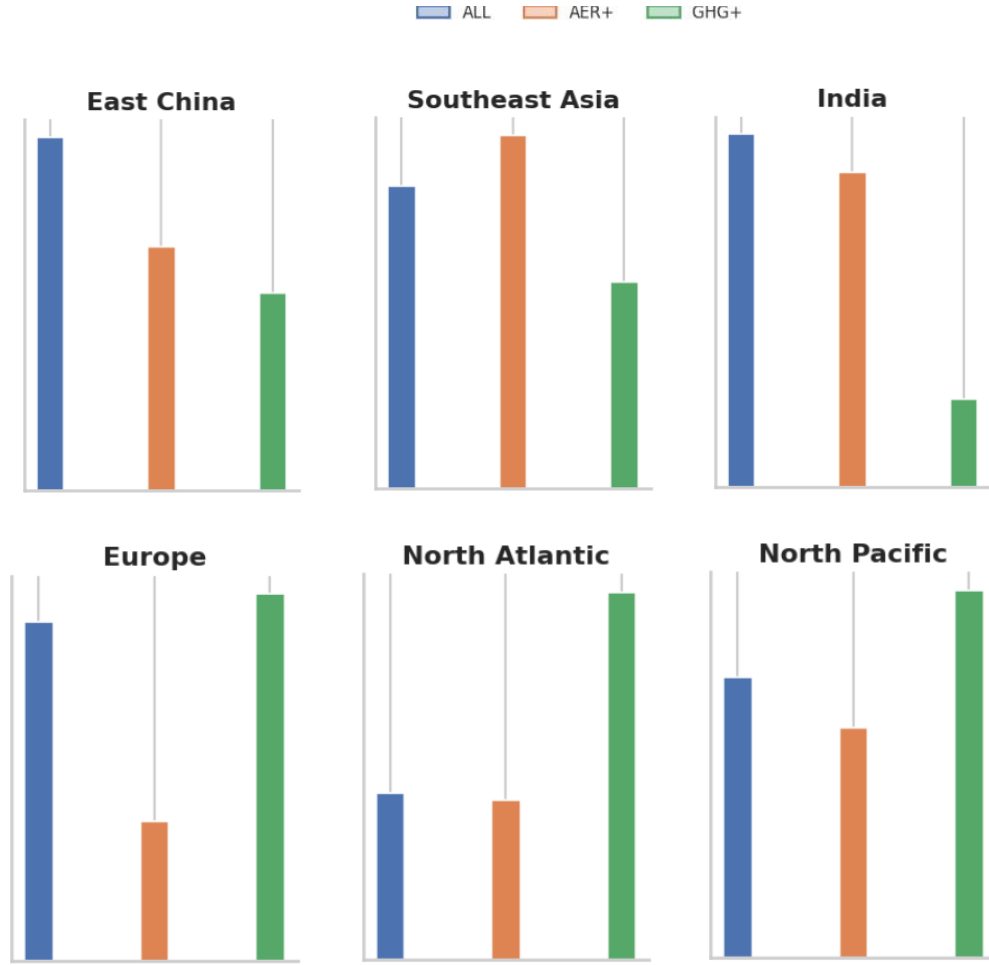


Figure 4.5. Histograms of mean relevance from layer-wise relevance propagation (LRP) over East China (a; 20°N–45°N and 100°E–125°E), Southeast Asia (b; 10°S–20°N and 95°E–130°E), India (c; 8°N–37°N and 68°E–97°E), Europe (d; 40°N–65°N and 0°E–40°E), the North Atlantic (e; 30°N–65°N and 80°W–0°W), and the North Pacific (f; 0°N–60°N and 120°E–120°W). LRP values are averaged over each year from 1920 to 2019.

GHG+ cases, the relevance is relatively diffuse and scattered, with weak contributions coming from parts of the North Atlantic, Eurasia, and the Indo-Pacific region.

Whereas, the AER+ relevance map shows more extreme values, with localized regions giving stronger positive and negative relevance. However, these signals appear less spatially coherent and more fragmented, making them difficult to interpret physically.

From the above figure, the average relevance values obtained using Layer-wise Relevance Propagation (LRP) for different important climate regions, including East China, Southeast Asia, India, Europe, the North Atlantic, and the North Pacific. The graphs compare the contribution of these regions under three different forcing experiments: ALL, AER+, and GHG+.

From the figure, it can be observed that different regions contribute differently depending on the forcing scenario used for training the ANN model. In East China and India, the ALL experiment shows comparatively higher relevance values, indicating that these regions play an important role when all climate forcings are considered together. Southeast Asia shows stronger relevance in the AER+ experiment, which may be linked to the influence of industrial aerosol emissions in that region.

Europe, the North Atlantic, and the North Pacific show higher relevance values in the GHG+ experiment. This suggests that greenhouse gas forcing has a stronger influence on the temperature patterns learned by the ANN in these areas. In particular, the North Atlantic and North Pacific regions appear consistently important across multiple experiments, indicating their strong connection with large-scale climate variability and long-term warming trends.

The relevance distributions demonstrate that the ANN is able to identify meaningful regional climate signals from the dataset. The LRP analysis helps in understanding which geographical regions contribute most strongly to the model's predictions under different climate forcing conditions.

Overall, while some localized features can be identified, the lack of clear large-scale structure indicates that the model finds it more challenging to extract stable and interpretable patterns for this period, especially for aerosol-driven variability.

We can check the contribution from some specific regions that are learnt by the ANN from the period of 100 years starting from 1920. It is easy to notice that the contribution ,in GHG+ experiment, are coming strongly from regions of North Atlantic, North Pacific and Europe. If we go by the historical significance of industrialization in these regions, these graphs show a related story. The North Atlantic region's GHG+ relevance maybe due to the Warming Hole, that show an opposite cooling trend.

In the later decades of last century, China and SouthEast Asia were significantly emitting industrial aerosols, which may be the reason for high relevance evident in the bar corresponding to AER+.

CONCLUSION

Due to complex interactions between internal and external forcings in the climate system, it remains difficult to estimate the local and regional influence of human-induced climate change on surface air temperatures. Our work shows the utility of XAI methods for finding patterns of climate signals because of changing external forcing. By utilizing an XAI tool as a pattern recognition method, we aim to understand how a nonlinear ANN makes a prediction by learning regional climate signals.

We found that our ANN trained on a large ensemble with the most realistic external forcing configuration (ALL) made predictions of real world temperature observations that correlated higher with the actual year. The LRP maps based on observations indicate that the temperature signal in the North Atlantic is particularly relevant for the predictions by the ANN trained on GHG+ compared to ALL. We also find good relevance in regions of East China and SouthEast Asia that may relate to the emission of industrial aerosols in later decades of 20th Century.

However, finding specific geophysical and historical relations that helped the ANN to learn temporal evolution is still a cumbersome task, and our work does not stand up to this task. All in all, the LRP implementation do work out and give satisfactory performance in context to our aim of exploring XAI methods for geoscience applications.

We further would like to delve into variations of LRP implementations, and notice how the relevance maps turn out to be.

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